

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

COURSE NAME: Deep Learning

COURSE CODE: MLA0405

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T

Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).

To satisfy the **Level 4 – Exemplary** rubric, write the answer like this:

Answer

Given:

Total number of coin tosses, ($n = 10$)

Outcomes: **H H T H H H T H H T**

Number of Heads, ($H = 7$)

Number of Tails, ($T = 3$)

Method (Maximum Likelihood Estimation)

For a Bernoulli process (coin toss), the likelihood function is

$$L(p) = p^H(1 - p)^T$$

where:

p = probability of getting Heads,

$H = 7$,

$T = 3$.

The Maximum Likelihood Estimator (MLE) for the probability of Heads is

$$\hat{p} = \frac{H}{H + T}$$

Calculation

$$\hat{p} = \frac{7}{10} = 0.7$$

Result

The estimated probability of getting **Heads** is

$$\boxed{\hat{p} = 0.7}$$

The estimated probability of getting **Tails** is

$$1 - \hat{p} = 1 - 0.7 = 0.3$$

Interpretation

The MLE estimates that the coin has a **70% probability of producing Heads** based on the observed data. This estimate best explains the given outcomes and can be used to predict future tosses of the coin.

2. Out of 50 students, 42 passed an exam.

Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Answer

Given:

Total number of students, $n = 50$

Number of students who passed, $x = 42$

Number of students who failed, $50 - 42 = 8$

Method (Maximum Likelihood Estimation)

Assuming a Bernoulli distribution, the likelihood function is

$$L(p) = p^{42}(1 - p)^8$$

where p is the probability that a student passes.

The Maximum Likelihood Estimator (MLE) is

$$\hat{p} = \frac{x}{n}$$

Calculation

$$\hat{p} = \frac{42}{50} = 0.84$$

Result

Estimated probability of passing:

$$\boxed{\hat{p} = 0.84}$$

Estimated probability of failing:

$$1 - \hat{p} = 1 - 0.84 = 0.16$$

Interpretation

The MLE estimates that a student has an **84% probability of passing** the exam based on the observed data, while the estimated probability of failing is **16%**.

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

Answer

Given:

Total number of bulbs, $n = 100$

Number of defective bulbs, $x = 8$

Number of non-defective bulbs, $100 - 8 = 92$

Method (Maximum Likelihood Estimation)

Assuming a Bernoulli distribution, the likelihood function is

$$L(p) = p^8(1 - p)^{92}$$

where p is the probability that a bulb is defective.

The Maximum Likelihood Estimator (MLE) is

$$\hat{p} = \frac{x}{n}$$

Calculation

$$\hat{p} = \frac{8}{100} = 0.08$$

Result

Estimated probability of a bulb being defective:

$$\boxed{\hat{p} = 0.08}$$

Estimated probability of a bulb being non-defective:

$$1 - \hat{p} = 1 - 0.08 = 0.92$$

Interpretation

The MLE estimates that the probability of a bulb being **defective is 0.08 (8%)**, while the probability of a bulb being **non-defective is 0.92 (92%)** based on the observed sample.

4. A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Answer

Given:

Total number of die rolls, $n = 60$

Number of times 6 appears, $x = 18$

Number of times 6 does not appear, $60 - 18 = 42$

Method (Maximum Likelihood Estimation)

Treat rolling a **6** as a Bernoulli event.

The likelihood function is

$$L(p) = p^{18}(1 - p)^{42}$$

where p is the probability of rolling a 6.

The Maximum Likelihood Estimator (MLE) is

$$\hat{p} = \frac{x}{n}$$

Calculation

$$\hat{p} = \frac{18}{60} = 0.30$$

Result

Estimated probability of rolling a 6:

$$\boxed{\hat{p} = 0.30}$$

Estimated probability of not rolling a 6:

$$1 - \hat{p} = 1 - 0.30 = 0.70$$

Interpretation

The MLE estimates that the probability of rolling a **6 is 0.30 (30%)**, while the probability of **not rolling a 6 is 0.70 (70%)** based on the observed data.

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.
2. Calculate the error.
3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Answer

Given:

Input, $x = 4$

Weight, $w = 2$

Actual Output = 10

Learning Rate, $\alpha = 0.01$

Gradient = 4

1. Predicted Output

Prediction:

$$\begin{aligned} y &= w \times x \\ y &= 2 \times 4 = 8 \end{aligned}$$

Predicted Output = $\boxed{8}$

2. Calculate the Error

$$\begin{aligned} \text{Error} &= \text{Actual Output} - \text{Predicted Output} \\ &= 10 - 8 = 2 \end{aligned}$$

Error = 2

3. Update the Weight

Gradient Descent update rule:

$$w_{\text{new}} = w - \alpha \times \text{Gradient}$$

$$w_{\text{new}} = 2 - (0.01 \times 4)$$

$$w_{\text{new}} = 2 - 0.04 = 1.96$$

Updated Weight = 1.96

Result:

Predicted Output = 8

Error = 2

Updated Weight = 1.96

Code of Conduct:

*I **Yoga Madha A-192425058** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation